

Proximities & Perspectives: Hybrid Nail Art Explorations Through a Research Lab-Situated Artist Residency

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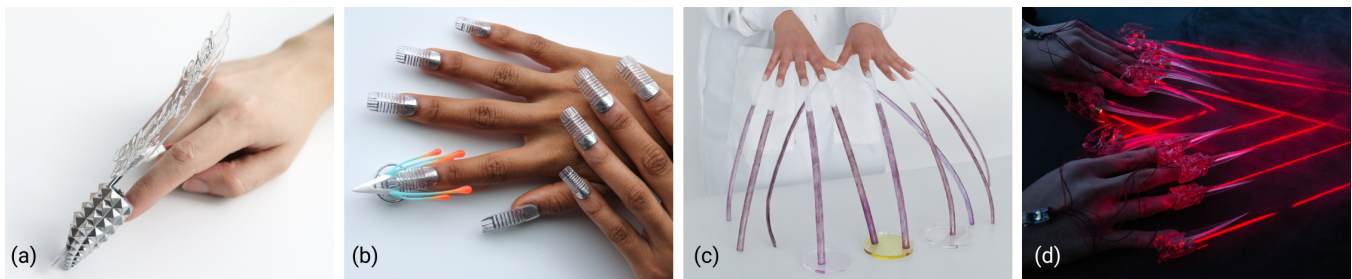


Figure 1: Hybrid Nail Art Case Studies: (a) Nail Tattoo Machine; (b) Gossip Nail; (c) Xtreme Long pH Sensing Nail; (d) Laser Nail

Abstract

Nail art has long been a form of personal expression and a significant part of the beauty industry. More recently, it has emerged as an intimate interface within the fields of Wearable Computing and Human-Computer Interaction. However, there is a lack of perspectives from nail artists, who bring rich design experience and unique perspectives on on-body aesthetics. In this paper, we present a series of hybrid nail art explorations developed through an artist residency situated in a research lab. Through four case studies, we demonstrate a range of interactive possibilities enabled by hybrid nail art, co-created by two nail and body artists in collaboration with wearable technology researchers. These works illustrate how nail art can serve as an extension of the body, enabling experiences across different proximities and perspectives. Lastly, we reflect on this collaboration and reveal insights on future opportunities for hybrid nail art.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI).**

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Keywords

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1 Introduction

The desire to modify one's physical appearance—*body crafting*—has been evident throughout human culture and history [19]. Nail art is a body crafting practice that focuses on modifying the appearance of one's fingernails. Examples such as nail extensions, nail art stickers, and nail polish have not only become a daily habit for many individuals, but have also contributed to the \$570 billion beauty industry [3].

In the past decade, since 2015, there has been an emergence of *hybrid nail art* in the Wearable Computing and Human-Computer Interaction (HCI) community. This is enabled in part by the miniaturization of electronics and innovations in novel materials, which allow technology to be integrated directly onto the nail surface [16, 24, 28]. However, these devices are often developed with a dominant focus on engineering. There is a lack of perspectives from nail artists and nail technicians, who hold expert insight into on-body design.

In this paper, we present an experimental approach to bridge technical and on-body design perspectives for the creation of hybrid nail art. Specifically, we adopted the approach of an Artist-in-Residence (AiR) situated in an HCI research laboratory as a research vehicle to explore alternative perspectives for hybrid nail art development beyond engineering. We present our process in leveraging an AiR as a research vehicle, the implementation, and fabrication of the resulting 4 case studies in which technology is leveraged as an expressive medium (instead of only a functional element) for nail art (Figure 1). We also share reflections on this approach and the opportunities for more inclusive designs in hybrid nail art.

2 Related Work

Nail Art. The first nail paints, used to signal social status, appeared in ancient Egypt, made from materials such as plant-derived henna and crushed beetle dye [18, 31]. In ancient Chinese culture, men and women of high social status would grow extremely long nails, which they would protect with nail guards [18]. In modern culture, many techniques for nail art have been introduced via the Beauty Industry [3]. The advent of nitrocellulose in the twentieth century enabled the creation of durable and widely commercialized nail polish [31]. Nail salons are a booming business offering various services, including attaching nail extensions and applying gel-based nail art, which solidifies under UV light for a longer-lasting polish. The tensions of labor and sexuality aroused in nail art has also been explored in gender studies [22, 26].

Hybrid Nail Art in HCI. In HCI and wearable computing, craft-based wearable design [12], expressive fashion technology [1, 8, 10], and speculative design frameworks [17] have explored the intersection of technology, aesthetics, and on-body art. Specifically, researchers have investigated nail-mounted devices and also bio-based materials and chemical-based sensing in a nail form:

- **Nail-mounted devices.** HCI researchers have explored nail-mounted outputs such as small OLED [28] and e-INK displays [16], and also haptic arrays [20, 32]. For inputs, nail-mounted capacitive touch input [24], nail-mounted cameras [52] were demonstrated. The FingerPad project [13] presented hall effect sensors to turn the fingertip into a touchpad. For wireless communication, Vega demonstrated nail-extensions integrated with RFIDs [34].

- **Bio-based and chemical-sensing nails.** HCI research has investigated biodegradable materials for artificial nails towards more sustainable manicures. Bio-E-Nails [11, 33] presents a bio-plastic formulation and end-of-life strategy to consider the entire life cycle of the fingernail. Fingernails chemically embedded with pH sensing capabilities have also been examined [25].

Artist-in-Residence (AiR) as a Research Approach. There is a growing interest in hosting AiRs in HCI research laboratories [15] as an experimental approach for interdisciplinary research between STEM and the Arts. This also builds on prior residencies and collaboration models from the tech industry: tech companies including Microsoft [6], Google [5], Autodesk [2], and OpenAI [7] have developed AiR programs to invite new perspectives towards tech. HCI academic labs have similarly involved artisans for digital fabrication [21], craftspeople for e-textiles [14], and artists for HCI topics [4, 9]. However, this has not been extended to the realms of nail art research, to which this paper makes a contribution.

3 AiR as Research Approach for Hybrid Nail Art

Here, we describe our approach in leveraging an AiR as a research vehicle. In this paper, we refer to the nail and body artists as "artists" and the wearable computing researchers as "researchers"; however, all the artists and researchers have contributed to the artistic exploration and research contributions collaboratively.

Open Call: The organizing researchers planned an 8-week residency program open call targeted for body artists, including nail, tattoo, and hair art. Prior experience in technology is not required. The open call was shared across social media platforms, mailing lists, and the lab website. The open call received 40 applications.

Selection Process: Three board of advisors were invited to guide the selection process, with backgrounds in beauty culture, wearable art, and apparel design. Together, the board of advisors and organizing researchers reviewed applications based on the portfolio, residency proposal, and curriculum vitae. Four finalists were interviewed online, and one artist duo (consisting of a nail artist and a tattoo artist) was selected based on the depth of their work, the fit of their proposal, and their openness to collaboration.

8-Week AiR in HCI Research Lab:

Week 1: Skill Exchange: The researchers hosted a series of hands-on, mutual skill exchange workshops for the selected artist duo to share their nail and tattoo practices, and for researchers to introduce possible techniques to enable hybrid functions. The researchers also provided a hands-on tutorial on fabricating on-skin interfaces, which the artists completed.

Week 2-7: Project Co-Creation: The artists led the project ideation, guided by their artistic vision for hybrid nail art. Researchers engaged in daily discussions with the artists to explore ideas, prototyping methods, and potential research contributions. The projects drew inspiration from academic research as well as references from nail art, fashion, and beauty practices. Multiple researchers collaborated on specific projects, providing expertise in areas such as miniaturized electronic prototyping, chemical engineering, material synthesis, and mechanical engineering.

Week 8: Presenting Residency Outcome: The resulting projects are presented through an artist talk delivered by both artists, shared on the artists' social media, and documented for later submission as presented in the Case Studies section of this paper. The open-source tutorials of the fabrication processes are also documented, providing public access to practitioners.

4 Case Studies

We present a case series of hybrid nail art created through the AiR research approach.

4.1 Nail Tattoo Machine

The nail tattoo machine is designed to integrate a tattoo machine into a nail cover form factor, enabling the direct application of tattoos onto the skin surface (Figure 2). Tattoos serve as a medium for marking the present and creating memories. The nail tattoo machine design draws inspiration from the traditional elegance of feather pens, which were once used to document. This case study explores the combination of on-body design tools, such as a tattoo machine and on-body stylus [27], with hybrid nail art. It extends the interaction modality to in-body interactions.

Electronic Integration: The tattoo machine function is enabled by a miniaturized solenoid attached to a tattoo needle. We used a TIP120 transistor to drive the solenoid in linear action, and used a potentiometer to adjust the frequency. The controller circuit is soldered onto a perfboard and assembled inside a 3D-printed enclosure. It can be powered either through a USB cable or using a standard tattoo machine power supply, which also comes with a foot pedal control commonly used by tattoo artists during the practice. The function of the tattooing machine is tested on tattoo practice skins with regular tattoo ink.

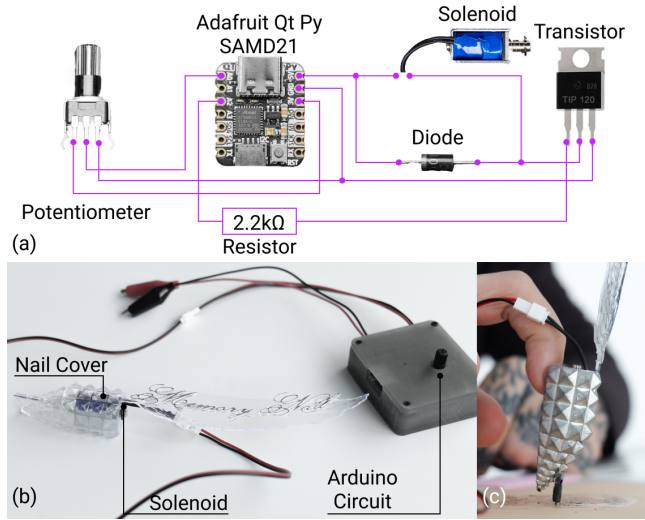


Figure 2: Nail Tattoo Machine: (a) Schematic; (b) Main components; (c) Testing on tattoo practice skin.

4.2 Gossip Nail

The gossip nail aims to extend the connection between people through a microphone nail and a speaker ring that serve as a pair of communication devices (Figure 3). The visual elements of the nail cover design were inspired by retro microphone covers and face-mounted microphones. The unique placement on nails and fingers enabled new experiences of speaking and listening.

Electronic Integration: The microphone function is enabled by an electret microphone amplifier MAX4466 with adjustable gain, and the speaker function is enabled by a 1-watt mini valve speaker. As a proof-of-concept prototype, we pre-recorded sound into the microcontroller for the speaker and used the onboard LED to visualize the volume of the sound captured by the microphone. In future work, this interaction can be enhanced through wireless communication to establish real-time communications.

4.3 Laser Nail

The Laser Nail explores the extension of personal space through elongation with laser beams (Figure 4). Each nail is seamlessly integrated with a red laser diode, visually extending the length of the nails beyond the nail structures themselves. The sharp, pointy nail design deliberately reflects the project's purpose to enable precise pointing, merging the utilitarian function of a laser pointer with the distinctive aesthetic of laser light beams.

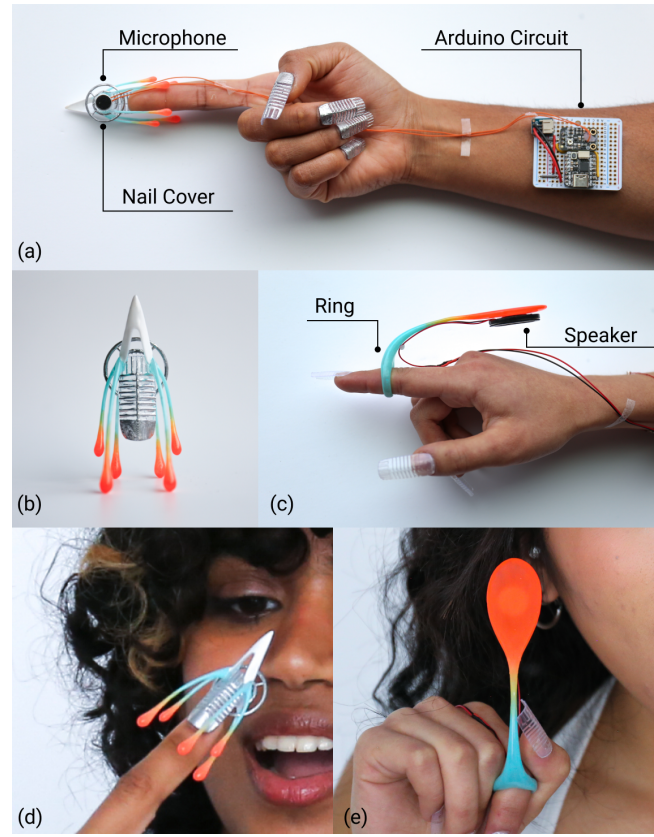


Figure 3: Gossip Nail: (a) Microphone nail components; (b) Nail cover front view; (c) Speaker ring main components; (d) Microphone nail in use; (e) Speaker ring in use.

Electronic Integration: We soldered five laser diodes to each battery connector with a switch, and used two LiPo batteries to power each of the diode groups. The peak wavelength of the laser is 650 nm, which is considered safe for brief exposure.

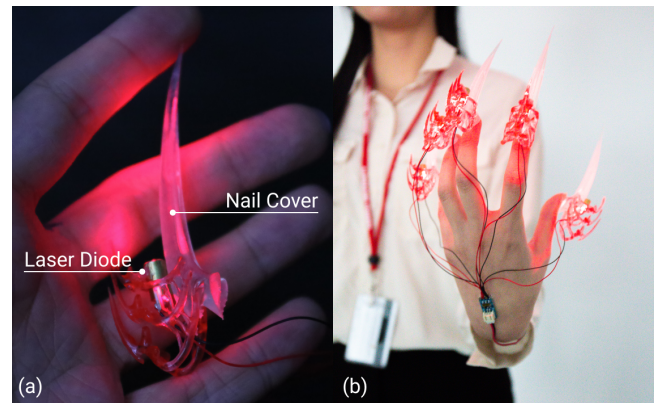


Figure 4: Laser Nail: (a) Laser nail main components; (b) Laser nail full hand view.

4.4 Xtreme Long pH Sensing Nail

Extremely long nails change color according to pH of the surrounding environment. Inspired by the Guinness World Record holders for the longest nails, these extremely long nails, measuring approximately 50 cm, can change color when the nails are submersed in liquid or soil. The nails respond to changes in pH levels, shifting from pink hues in acidic solutions to blue hues in alkaline solutions.

Fabrication Implementation: Each of the 50 cm long press-on nail is divided into four segments to fit into a regular Formlabs 3 resin 3D printer. After printing and support removal, each segments are connected through pre-designed notches, then adhered using UV resin on the inner side of the nails. Different from prior work on fingernail colorimetric chemical sensing using lab-grade chemicals [25], we adopted recipes from HCl colorimetric sensor fabrication [29, 30, 35], using red cabbage powder mixed in water-based clear nail polish to synthesize colorimetric nail polish at a 4% weight/volume (w/v) concentration. We applied 8 layers of the mixed nail polish onto the 3D printed nail, and waited for it to dry between each layer.

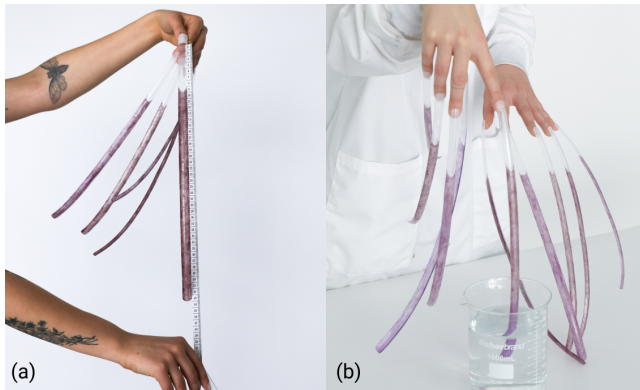


Figure 5: Xtreme Long pH Sensing Nail: (a) Measuring the length of the nail; (b) Testing the nail in a pH buffer solution.

4.5 Implementation

We synthesized the fabrication approach for creating hybrid nail art through a combination of 3D printing, nail polish painting, and interaction integration.

3D Resin Printing Nail Covers: The 3D models of the nail covers were created in Blender, a software that the artists are already familiar with. Formlabs Clear Resin V4 was used for all projects to give more flexibility with the colored designs. After printing, extra care was taken to clean the supports.

Coating and Curing: We used UV gel nail polish to coat the nail cover to achieve different colors and finishes, including chrome, opaque, and translucent colors.

Interaction Integration: Electronic components were integrated after the nail cover fabrication process and tested before being put on nails. To ensure that the nail covers are lightweight and wearable, the controller circuits were connected to on-nail components with thin, flexible wires.

Applying the Nail Cover: To apply the nail art onto the models, their natural nails were filed and cleaned with isopropyl alcohol.

Then, a clear, double-sided nail sticker was applied to the natural nail. The nail art was attached starting from a 45° angle from the nail tip to the nail bed to ensure proper adhesion.

5 Discussion and Conclusion

Nail art as an under-explored on-body design. The wearable technology researchers involved in this residency are more experienced in designing for common body locations such as the wrist and head. In contrast, the human nail has received less attention, including how to design for its curved shape and how to balance technical complexity with comfort. Unlike other areas in wearable design, there are no standard measuring or sizing techniques for nail design. Yet, fingernails offer rich opportunities as unobtrusive interactive surfaces. The wide range of shapes, lengths, materials, and finishes presents a valuable design space for future exploration.

Hybrid nail art as body extension for different proximity and perspectives. The series of hybrid nail art case studies in this work illustrates how nail art can serve as an extension of the body, enabling experiences across different distances and viewpoints, echoing Kao's Hybrid Body Craft framework on the communicative qualities of body crafts [23]. For example, the tattoo nail pierces into the skin, the gossip nail connects people over a distance, the laser nail extends the body's reach, and the pH nail senses the surrounding environment.

Challenges and opportunities in hybrid nail art. During the mutual skill exchange, the artists demonstrated their manual process of shaping three-dimensional nail art using acrylic powder. However, the process requires refined sculpting techniques that are difficult to learn and apply. Leveraging 3D modeling and 3D printing, the artists were able to speed up the iterative prototyping process. The nails can be printed in batches and produced efficiently when testing colors and finishes. However, each nail cover still requires careful measuring and model adjustments to ensure a proper fit. In addition, limited access to facilities remains challenging for nail artists. Currently, few practicing nail artists use 3D printing as their primary method due to these limitations.

Reflections on artist residency collaboration. As artist residencies have a way to explore multidisciplinary collaboration, we found this particular residency with nail artists highlights the contrast between hands-on techniques and digital fabrications. In digital fabrication research within HCI, new techniques are often introduced with a focus on being easy for beginners to learn. However, this emphasis can overlook the value of skilled craftsmanship. We observed that the unique combination of digital fabrication with refined sculpting and painting techniques in nail art brings important insights into how we understand craft and making practices. These insights could be further developed in future residencies. Moreover, there are very few residency programs focused on on-body art such as nails, tattoos, and makeup. These practices are often seen as services rather than art forms. This residency helped reaffirm the importance of recognizing nail art as a form of artistic practice.

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